Binary Activation in Transformers

FFN Evidence and a Modest Path Beyond Shared Attention

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Abstract

Transformers contain feed-forward networks (FFNs), but they are not merely FFNs. Attention projections, output mixing, residual streams, normalization, embeddings, caches, and output heads create paths outside a gate placed on an intermediate FFN activation. We therefore separate a demonstrated result from an open one. DistilBERT succeeds because its classifier and FFN surfaces expose constructible coordinate supports; the result does not depend on decomposing attention. In strict classification, binary activation placed before every FFN down projection gives exact confinement of the aligned trainable FFN parameters and optimizer state under a frozen shared backbone. Owning accuracy decreases from 91.28% at $R = 1$ to 86.21% at $R = 16$ as each regime's FFN slice narrows, while all measured unauthorized state remains bit-exact unchanged. Private residual adapters or experts recover much of the utility at additional per-regime storage cost.

This is not complete Transformer isolation. Shared attention remains shared, and naively ablating head projections disrupts a network that was not trained for those zeros. Standard self-attention is relational and input-dependent: Q/K dot products define token adjacency, softmax couples candidates, and output, normalization, and residual paths recombine the result. It does not present a canonical set of tenant constituents to which fixed supports can be assigned. A complete attention-lane construction would need separate Q/K/V and output projections, normalization, FFNs, and every residual path, with gates before recombination. That construction is expensive, vulnerable to operator error, and not established here. For current large language models we recommend the useful surfaces that do not depend on that open result: atomic whole-model Binary Activation, Cargo-authorized context, and explicitly gated external adapters or tools. Complete Transformer partitioning remains modest future work rather than a premise of the system. Mixture-of-Experts routing is a separate promising surface: authorization can restrict which already-discrete experts a principal may reach without claiming that shared attention itself has been partitioned. An $R = 8$ held-out experiment below trains a top-1 semantic router inside each fixed authorized expert set and then packs the trained lanes with exactly zero prediction change.

1 Scope

DistilBERT is a Transformer. Successful gating experiments on DistilBERT do not show that every part of a Transformer has been partitioned; they show that its FFN sublayers can expose a useful and testable gate surface. This distinction resolves an apparent contradiction: CDP can work *within* a Transformer without claiming that the complete Transformer is internally isolated.

We use *Binary Activation* for the operation

$$\tilde{\mathbf{h}} = \mathbf{h} \odot \mathbf{g}_r, \qquad \mathbf{g}_r \in \{0, 1\}^{d_{\mathrm{ff}}}. \tag{1}$$

At a declared FFN boundary, an active coordinate passes and an inactive coordinate is exactly zero. The credential-to-regime runtime decides which mask may be used. The result concerns that tensor and its aligned trainable state; it does not retroactively make the shared residual stream private.

The word *partition* carries a stronger requirement than the availability of multiplication. Binary Activation requires a constructible constituent space: stable indexed pieces, aligned parameters and optimizer state, and closure of every route by which an excluded piece could mix into or bypass the active support. The FFN expansion supplies such pieces locally. A standard pretrained attention computation does not supply tenant constituents in that form.

The paper answers four questions:

1. What exactly was gated in the successful classification experiment?
2. Does each regime receive only a fraction of FFN capacity?
3. Why does the same construction not isolate attention heads?
4. Which Transformer and LLM uses remain valuable now?

2 The Transformer Boundary

2.1 What the FFN gate covers

For one Transformer FFN sublayer, write

$$\mathbf{h} = \sigma(\mathbf{x}\mathbf{W}_1 + \mathbf{b}_1), \tag{2}$$

$$\tilde{\mathbf{h}} = \mathbf{h} \odot \mathbf{g}_r, \tag{3}$$

$$\mathbf{f}_r = \tilde{\mathbf{h}}\mathbf{W}_2 + \mathbf{b}_2. \tag{4}$$

The gate is on the expanded, post-nonlinearity activation immediately before the down projection. It is sometimes convenient to describe a related gate as *pre-MLP*: a mask applied at the input to an otherwise governed MLP. Both are FFN surfaces, but they protect different tensors and must be named precisely. The strict experiment in this paper uses the intermediate expanded activation in (4).

For an inactive hidden coordinate j, both its forward contribution and loss gradient at the gate are zero. With an optimizer that masks the complete update, including moment and weight-decay state, training is confined to the aligned columns of W_1, hidden-bias entries, and rows of W_2 in the mathematical row-vector convention.

2.2 What remains shared

A standard Transformer block also contains attention, residual additions, and normalization. In the evaluated construction those components are frozen and shared. They may compute common features for every regime and transport those features around the FFN branch. Consequently:

- the experiment proves exact inactive FFN parameter-state preservation;
- it does not prove that shared attention activations are independent;
- it does not assign private attention-head knowledge to each regime; and
- caches, embeddings, output heads, logs, and the serving runtime remain outside the local FFN theorem.

This is a declared trust boundary, not a hidden failure condition. It is useful when tenant-private trainable state is restricted to the gated FFN surface or a private lane, while the frozen backbone supplies public shared features.

2.3 Why classification still succeeds

There are two successful DistilBERT constructions, and they support different claims. In the five-seed formative study, the gate is applied to the final 768-dimensional CLS representation and the complete encoder is co-trained with the masks. Optimization can place discriminative features onto each retained support; the observed gated-versus-separate cost is 0.14–0.38 percentage points for $R = 8$–128. Because all regimes update shared encoder weights, this is a support-constrained utility result, not a tenant-state-confinement result.

In the strict construction, the shared attention backbone is frozen. The expanded FFN activation after the element-wise nonlinearity is an explicit coordinate tensor, and the adjacent projection slices and optimizer moments can be aligned with the same support. The model therefore retains shared Transformer feature extraction while each regime adapts only its governed FFN slices and private classifier. Classification remains useful because the task does not require the gate to manufacture independent attention heads; it only requires the governed downstream surface to separate the labels.

The tradeoff is correspondingly clear. Co-training gives the optimizer the most freedom but shares learned encoder state. Freezing the backbone and dividing a fixed FFN gives exact measured inactive-state preservation but reduces each regime to d_{ff}/R coordinates. Private adapters or experts restore capacity at linear per-regime storage cost. None of these results should be discarded; they answer different questions.

Replay formative classification. Run `experiments/reproduce_distilbert_classification.py` with `-surface legacy-post-encoder`; the five-seed artifact is `experiments/results/series1_results_20260803_202919.json`.

3 Strict FFN Classification Evidence

3.1 Construction

The experiment uses DistilBERT-base on AG News. Embeddings, attention, normalization, residual/shared parameters, and all other non-governed weights are frozen. A regime gate is installed on the 3,072-dimensional expanded activation before every FFN down projection. Only the active FFN slices and a regime-private classifier are trainable. Regime-scoped Adam moments and update steps are restricted to the same slices.

The model is trained on 8,000 examples for two epochs and evaluated on 2,000 held-out examples at seeds 42, 123, and 256. Each regime receives d_{ff}/R coordinates because the tested values divide 3,072. Thus the construction does trade per-regime FFN width for co-tenancy. It does not divide the width of the frozen attention backbone.

Table 1: Strict intermediate-FFN Binary Activation. Utility is measured; inactive-state preservation is checked exactly.

R	FFN dims/regime/layer	Owning accuracy	Drop (pp)	Max unauthorized delta
1	3,072	$91.28\% \pm 0.33\%$	0.00	0
2	1,536	$90.15\% \pm 0.39\%$	1.13	0
4	768	$88.93\% \pm 0.30\%$	2.36	0
8	384	$87.47\% \pm 0.04\%$	3.82	0
16	192	$86.21\% \pm 0.08\%$	5.07	0

The unauthorized-delta check covers frozen shared parameters, off-partition FFN parameters, off-partition first and second Adam moments, and inactive private classifiers. Every checked value is bit-exact unchanged. The table does not claim accuracy parity. It shows a capacity curve: narrower private FFNs retain useful classification performance while respecting an exact measured state boundary.

Replay strict FFN classification. Run `experiments/modal_dense_ffn_cotenancy.py`; retained runs are `experiments/results/dense_ffn_cotenancy_*.json` and the aggregate is `experiments/results/transformer_cotenancy_summary.json`.

3.2 Preliminary public mask-aware adaptation

The strict sweep above applies masks after generic pretraining. A separate $R = 8$ study asks whether the shared backbone can first learn the partition geometry on public data, before any tenant-stage training. At seeds 42, 123, and 256, an ungated teacher is fine-tuned for two epochs on 8,000 public AG News examples. A copied student then receives one additional public epoch in which every batch is evaluated through all eight 384-coordinate FFN masks with an equal mixture of hard-label and temperature-2 distillation loss. A disjoint 8,000-example split is subsequently used for strict tenant-stage training with all shared paths frozen.

The no-extra-training initialization reaches $88.55\% \pm 0.11\%$ owning accuracy after the tenant stage; the all-mask distillation initialization reaches $90.15\% \pm 0.11\%$, a difference of 1.60 ± 0.10 percentage points. The ungated teacher reaches $91.62\% \pm 0.35\%$, leaving the complete mask-aware pipeline 1.47 points below it. Frozen shared weights, off-partition parameters and optimizer moments, and inactive classifiers remain bit-exact unchanged during tenant training.

This is preliminary pipeline evidence, not a causal estimate. The baseline does not receive the student's additional public epoch, so the comparison cannot separate extra training, mask-aware adaptation, distillation, or their interaction. The public and tenant labels denote disjoint splits of one public benchmark rather than a real private-corpus transfer.

Replay status. The retained three-seed artifact is `experiments/results/public_gate_distillation_20260806T193105_795325Z.json`. The exact historical runner is not present. Its stronger equal-work successor is `experiments/modal_public_gate_adaptation_factorial.py`; the completed one-seed artifact is reported below and does not replace this historical three-seed pipeline comparison.

3.3 Private lanes

The capacity loss can be moved from shared width to per-regime storage. In an $R = 4$ frozen-backbone experiment, a private 64-dimensional residual adapter per regime reaches $90.47\% \pm 0.18\%$ owning accuracy and $3.18\% \pm 0.06\%$ wrong-key accuracy, with approximately 0.59 million parameters per regime. A private 3,072-dimensional expert reaches $90.09\% \pm 0.14\%$ owning accuracy and $3.30\% \pm 0.05\%$ wrong-key accuracy, with approximately 28.33 million parameters per regime. The measured unauthorized parameter delta is zero in both constructions.

The current-library seed-42 rerun gives 90.66% adapter and 90.15% expert owning accuracy, with 3.11% and 3.28% wrong-key accuracy. Shared-backbone and inactive-lane deltas are exactly zero, and malformed execution authority makes zero model calls. These one-seed values corroborate, rather than replace, the multi-seed estimates.

These lanes preserve shared attention while giving each regime separate trainable capacity selected by Binary Activation. They therefore resemble an identity-bound attachment architecture more than complete internal Transformer partitioning. The tradeoff is explicit: near-baseline utility is purchased with private parameters rather than obtained from a narrow disjoint slice for free.

3.4 Gate-aware public adaptation

A matched $R = 8$ factorial asks whether public training should expose every FFN support before the later tenant stage. The no-extra condition, an extra ungated hard-label condition, an all-mask hard-label condition, and an all-mask hard-label-plus-distillation condition use matched examples, batch order, dropout schedule, optimizer steps, teacher calls, and student forward/backward counts. In one full seed, ordinary extra training changes tenant accuracy by +0.075 percentage points, mask awareness adds +1.331 points beyond that control, and distillation adds +0.031 points given masks. The complete pipeline is +1.438 points above no extra adaptation. All frozen, off-partition, optimizer-moment, and inactive-head deltas are exactly zero. The reduced-data

pilot is negative; it is retained as evidence that a tiny optimization smoke is not a utility estimate. One full seed is still insufficient for a population-level effect claim.
Replay factorial. Run `experiments/modal_public_gate_adaptation_factorial.py`; the completed artifact is `experiments/results/public_gate_adaptation_factorial_20260821T063703_914999Z.json`.

4 Why Naive Attention Gating Fails

The obstacle is not that an attention tensor cannot be multiplied by zero. The obstacle is that a fixed binary support presupposes constituent pieces that the trained attention computation does not canonically provide. Self-attention represents relationships among tokens and distributed features rather than a stable list of tenant-owned semantic coordinates.

Multi-head attention is not a collection of independent outputs once a normal Transformer has been trained. For head h,

$$\mathrm{head}_h(X) = \mathrm{softmax}\left(\frac{XW_h^Q(XW_h^K)^\top}{\sqrt{d_h}}\right) XW_h^V, \tag{5}$$

and the concatenated heads are mixed by W^O, added to a residual stream, and normalized. Zeroing projections from a head after ordinary pretraining removes paths that the rest of the network expects. The resulting quality failure is not surprising evidence against the arithmetic $0 \cdot x = 0$; it is evidence that the remaining network was not trained for that ablation.

The affinity matrix is input-dependent. Q/K dot products encode vector norms and angular proximity; when norms are controlled, that angular term corresponds to cosine adjacency. A coordinate or head that is adjacent for one input need not define the same relationship for another. Changing Q, K, or a subset of their coordinates changes attention logits, and softmax renormalizes every candidate in the row. The effect is therefore global to that attention distribution rather than confined to one putative tenant constituent.

This is an ontological limit as well as a numerical one: fixed disjoint masks assume that the protected object can be constructed as separable pieces. A pretrained attention space supplies distributed, relational features, not a canonical tenant decomposition. Imposing a fixed tenant ontology on arbitrary head coordinates does not create the missing constituents.

More importantly, a gate on one attention projection does not close all paths:

- Q and K jointly determine attention weights, while V carries values; gating only one does not define an independent lane;
- W^O recombines all heads into the shared residual width;
- residual additions preserve an ungated route around a gated branch;
- LayerNorm mixes statistics across coordinates; and
- later shared FFNs and attention blocks can recombine the result.

The practical conclusion is direct: a post-hoc head mask is neither a complete isolation construction nor a performance-preserving transformation. This limit does not invalidate a gate on a genuine FFN constituent elsewhere in the same Transformer.
Replay negative controls. The local placement/bypass checks are in `experiments/local_transformer_cotenancy_suite.py`; the exploratory head-lane stress runner is `experiments/modal_attention_lane_stress.py`. The latter is retained as a failure diagnostic, not positive isolation evidence.

5 A Complete Lane Construction Is Open

An attention-capable isolation construction is conceivable only if the architecture first constructs the missing constituents and closes the entire lane. Each regime would require its own Q/K/V projections, attention output projection, normalization state, FFN path, and *every* residual route. A Binary Activation gate would be required at each boundary before lanes recombine. Training would need to be gate-aware from initialization or a carefully controlled distillation stage.

This is materially different from copying an attention head. A missed residual or normalization path invalidates the complete-lane claim, and duplicated lanes multiply memory and compute. The operator-risk surface is large enough that a paper claiming complete Transformer co-tenancy would need a topology audit, exact wrong-lane activation and state checks at every layer, owning-positive and wrong-key-negative utility controls, and bypass tests in the actual serving runtime.

We do not make that claim here. Complete Transformer partitioning is a research direction beyond the demonstrated Binary Activation surfaces.

5.1 Causal FFN placement check

A separate one-seed TinyLlama 1.1B run tests the narrower FFN theorem in a causal model. At $R = 8$, each 5,632-dimensional expanded SwiGLU product is masked immediately before the down projection, leaving 704 coordinates per regime. The exact probe finds zero inactive gradients and parameter changes; the partitions are disjoint and complete, initialization is matched, and all malformed execution authorities are rejected before model execution. Utility is not preserved: mean gated token loss is 5.442 versus 4.230 for the matched ungated control. Owning canary loss is 3.013 versus 3.452 under wrong keys, but none of 16 owning canaries is generated exactly. Thus the causal result checks placement and confinement; it is explicitly not generative parity.

6 Permutation Conjugation: A Limited Result

A residual-basis permutation can preserve a trained Transformer's function if every residual-facing parameter and normalization component is conjugated consistently. In the historical DistilBERT sweep, accuracy remained 90.25% with zero measured gap for all tested permutations from $R = 4$ through $R = 128$, and for eight sampled regimes at $R = 1{,}024$. A current-library rerun through the source-visible research preflight evaluates every addressed regime at $R = 8$ and $R = 128$: baseline and all 136 regime evaluations have identical 91.421% accuracy. The maximum fp32 logit drift is 1.06×10^{-5} and changes no prediction.

This is a useful implementation and equivariance check, not an isolation result. Conjugation expresses the same function in a keyed basis; it does not create distinct tenant knowledge and does not make non-tenant activations zero. An exact zero arises only when the authorized basis is composed with an actual Binary Activation gate. That gated composition then inherits the same closure requirements described above.

Replay. Run `experiments/orthogonal_superposition_sweep.py`; the retained sweep and $R = 1{,}024$ outputs are indexed by `experiments/results/README.md`.

7 MoE Expert-Router Authorization

Mixture-of-Experts architectures already expose discrete FFN experts and a router that produces expert logits, normalized dispatch weights, and a top-k selection (Shazeer et al., 2017; Fedus et al., 2022). This creates a more natural authorization surface than attempting to partition a dense attention representation. A tenant + sub-id grant may define an allowed expert set $A \subseteq \{1, \dots, E\}$; the router must ensure that only experts in A can be selected or launched.

The placement detail is essential. Multiplying pre-softmax logits by a binary mask is not sufficient: an unauthorized zero logit can still receive positive softmax probability. One conforming construction restricts candidates to the authorized set A (equivalently, assigns $-\infty$ outside A), selects at most $k' = \min(k, |A|)$ finite entries, and refuses to dispatch any $-\infty$ entry. Another multiplies normalized dispatch weights by the grant mask before dispatch and removes every zero-weight entry. Both fail closed when A is empty. Setting logits to $-\infty$ without constraining a fixed-size top-k is insufficient when $|A| < k$. In the normalized-weight construction, unauthorized expert contributions are exactly zero by the same Hadamard algebra as the FFN gate; exclusion from selection and execution is additionally a router-implementation property.

MoE routing remains learned and data-dependent—it decides which authorized experts should process a token. Binary Activation is grant-scoped and data-independent—it decides which experts the principal may reach at all.

We evaluate that composition on eight disjoint binary tasks from the standard 20 Newsgroups train/test split. Each regime independently trains a top-1 router over two private feed-forward experts with task loss and a load-balancing loss computed only inside its authorized set. The eight trained routers and sixteen experts are then copied unchanged into one execution-authorized bank. Across 6,231 held-out examples, macro accuracy is 81.80% and micro accuracy is 81.86% (5,101 correct). Separate and packed execution have exactly equal logits and predictions; unauthorized expert dispatch is zero. Both experts are used in every regime, with the least-used expert receiving 42.4% of its regime's held-out examples, and every router records nonzero parameter movement. A dedicated active-regime training-step audit records zero inactive gradient tensors, optimizer states, and parameter change. Wrong regime, model, dimension, and empty-authority probes against eight single-regime execution surfaces make zero model calls.

The deliberately invalid control multiplies unauthorized pre-softmax logits by zero; an unauthorized expert wins top-1 because zero exceeds the authorized negative logits. Candidate restriction or $-\infty$ masking before softmax is therefore an empirically exercised placement requirement, not documentation alone. Exact zero dip here is a copy-and-pack equivalence result. The packed bank has the same router/expert bytes as the separate lanes, so this experiment does not itself claim storage compression, shared-attention isolation, or language-model quality.
Replay learned routing. Run `experiments/train_authorized_moe.py`; the clean-source artifact is `experiments/results/authorized_learned_moe_20260831T182431_055309Z.json`.

7.1 Execution-Selected Capability Prefixes and Token Routing

Learned continuous prefixes can condition model computation without changing the user-visible token sequence (Li and Liang, 2021). They are not credentials: a plaintext prefix supplied by a requester is forgeable and remains untrusted data. We separate those channels. Authenticated execution authority selects a private learned prefix and its two-expert candidate set; user tokens then condition top-1 routing only inside that set. The private prefix enters the token router as continuous context, and authorization is enforced before softmax and dispatch.

On the same eight held-out 20 Newsgroups task pairs, the experiment makes 311,477 non-padding token-routing decisions and obtains 77.76% macro and 77.80% micro accuracy (4,848 of 6,231 examples). Packing the independently trained prefixes, token routers, and experts changes zero routes, logits, or predictions. Global negative-infinity masking exactly matches candidate restriction, and unauthorized dispatch is zero. A spoof request containing all eight literal `CAPABILITY_REGIME_n` markers as user text causes zero unauthorized dispatch. Every prefix and router moves during training, both experts are used in every regime with minimum held-out share 41.7%, and the active-regime audit records zero inactive gradients, optimizer states, or parameter change. Malformed probes against all eight singleton execution authorities make zero model calls.

This is a small token-routing classifier, not an autoregressive Transformer. It establishes neither prompt text as authority nor generation quality, shared-attention isolation, or compression. Exact zero dip means only copy-and-pack equivalence; task utility is reported separately.
Replay capability-prefix routing. Run `experiments/train_capability_token_moe.py`; the clean-source artifact is `experiments/results/capability_prefix_token_moe_20260831T182453_013039Z.json`.

8 Useful LLM Surfaces Now

The absence of complete attention partitioning does not eliminate practical Transformer uses:

- **Whole-model Binary Activation.** Treat the complete LLM as an atomic capability. A tenant + sub-id grant can bind model/version, operation, expiry, token ceiling, use count, and audit context. Unauthorized requests are rejected before inference. Model quality is unchanged.
- **Cargo Mode.** Authorize a tenant corpus or retrieval partition before its contents become context for an ordinary LLM. A separate model gate can then authorize generation. This controls data release and invocation, not internal attention.
- **Gated attachments.** Select a private LoRA adapter, expert, reranker, database, or tool only after resolving the authenticated scope. The attachment retains its ordinary performance and storage characteristics; Binary Activation supplies the fail-closed authorization point.
- **FFN-private classification.** When a frozen shared Transformer is acceptable, the demonstrated FFN and private-lane constructions provide measured tenant-private trainable state with explicit capacity costs.
- **MoE expert authorization.** Restrict an existing expert router to a grant-scoped allowed set before dispatch. This controls expert reachability but does not make shared attention, caches, or residual state tenant-private.

These surfaces share one conceptual vocabulary—activate the authorized capability or do not execute it—while preserving honest differences in what each boundary isolates.

Replay authorization surfaces. Whole-model and local Cargo checks are in `experiments/local_transformer_cotenancy_suite.py`; the real-model Cargo runner is `experiments/modal_cargo_transformer_authorization.py`.

9 Research Protocol

If complete Transformer lanes are revisited, a decisive experiment should begin small and test the construction rather than raw post-hoc ablation:

1. train or distill a two-lane Transformer with separate Q/K/V, W^O, normalization, FFN, and all residual routes from the outset;
2. statically enumerate every cross-lane tensor edge and gate before any recombination;
3. train an owning-positive task in each lane while freezing or separately accounting for all shared state;
4. verify wrong-key outputs, activations, parameter deltas, optimizer moments, and caches; and
5. compare owning utility with a compute- and parameter-matched ungated control.

Failure would be informative, but success should not be presumed by the current FFN result. Until that protocol passes, the system should use whole-model, Cargo, attachment, or FFN-local wording rather than "Transformer isolation."

10 Conclusion

The Transformer result is narrower and more useful than either extreme. It is not true that Binary Activation has no place in Transformers: DistilBERT's FFNs and classifier representations provide demonstrated gate surfaces, private lanes offer an explicit storage–utility tradeoff, and whole-model or Cargo gates control LLM inference without touching attention. It is also not true that an FFN gate partitions the complete Transformer.

Shared attention is accepted as shared in the demonstrated construction. Complete lanes would require duplicated projections, normalization, FFNs, and all residual paths, with substantial compute and operator risk. That remains modest future work. The present recommendation is to deploy Binary Activation where its boundary is exact and auditable, and to state that boundary plainly. MoE expert authorization offers a narrower near-term Transformer path because the experts are already discrete. The held-out learned-router result validates the fail-closed candidate-restriction and lossless-packing mechanism in a small feed-forward MoE. The token-routing extension further shows that execution can select a private continuous router prefix while capability-looking user text remains ordinary data and cannot expand the expert set. Applying either result inside an autoregressive language-model MoE remains future work.

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